A Reproducibility and Transportability Audit of Cardiovascular Prediction Under Dataset Shift

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Abstract—This work audits whether a leakage-controlled, multi-source cardiovascular prediction workflow transports to unseen datasets. On the source-distinct, endpoint-aligned AngioCAD cohort, the primary source-only analysis achieved ROC-AUC 0.787 (95% conditional interval 0.734–0.838), while calibrated 10-bin expected calibration error was 0.216 and feature-only out-of-distribution AUROC was 0.589 (0.541–0.635). AngioCAD reported 413 labeled IDs, but only 377 had matching feature rows; the public workbook does not document age or fasting blood-sugar units. In a post-hoc replay, the integrated selective rule emitted 21/377 outputs (5.6%), all for positive cases, which is too small and one-class to establish useful risk control. A metadata-based site gate withheld scores for every unregistered site. Duke supplied a separate stress test with a different stenosis threshold. The evidence supports a reproducible engineering audit with negative transportability findings; it does not establish dependable calibration, useful unseen-site coverage, or feature-only shift recognition.

Index Terms—cardiovascular disease, leakage control, nested validation, explainable AI, calibration, conformal prediction, selective prediction, dataset shift, transportability.

I. Introduction

Cardiovascular prediction models must be evaluated for discrimination, calibration, robustness, and transportability rather than accuracy alone. CARDIODIAG provides an important starting point because it combines explainable feature reduction with optimized ensemble classifiers [1]. The active question is whether a model trained across source cohorts can preserve useful discrimination and probability reliability on unseen sites, and whether its inference policy can withhold scores outside its registered operating domain. The broader stability audit remains a complementary question. Selective reliability is measured empirically and is not treated as evidence that a model semantically understands uncertainty.

Contribution status. The supported contribution is a reproducible engineering evaluation of multi-source transfer, source-only uncertainty and calibration, feature-only shift diagnostics, and a fail-closed unknown-site policy, supplemented by a stability–reliability decoupling analysis. The evidence includes a negative result: stable feature selection did not yield a consistent reliability advantage, and feature-only shift detection was weak on the aligned external site. No new prediction algorithm is claimed. One external site does not establish broad generalization or clinical utility.

Author order and common affiliation are provisional and require confirmation by all authors before submission.

II. Related Work and Research Gap

Feature-selection stability is an established property of a selection procedure, not a substitute for predictive validation. Nogueira et al. provide a chance-corrected stability statistic for comparing selected sets [2], while resampling-based stability selection has distinct assumptions and error-control goals [3]. Nested validation is required when the same data support model selection and performance estimation [4], [5]. Calibration, discrimination, and selective risk are likewise distinct properties of a prediction model [6]. Conformal prediction can provide marginal coverage under exchangeability, but coverage should not be assumed under population shift [7].

The research question is whether training-only feature-set stability in the CARDIODIAG representation is associated with held-out predictive and selective reliability, and whether source-only uncertainty methods remain informative under dataset shift. This is an evaluation question, not a claim that the combination is unprecedented. The reporting and risk-of-bias framing follows TRIPOD+AI and PROBAST+AI [8], [9].

III. Audit of the Original Implementation

The supplied repository is notebook-centric and contains multiple historical protocols. The historical reproduction and its deviations are documented in docs/baseline_reproduction_report.md. In particular, available notebooks contain paths where resampling, preprocessing, and dimensionality reduction occur before the evaluation split.

A. Historical Reproduction Claims

The original paper's reported protocol cannot be reproduced bit-for-bit from the supplied notebooks: split sizes, preprocessing order, model blocks, search grids, and saved tables are inconsistent. The native reproduction path specified in the audit applied SMOTE before an 80/20 split without stratification; on heart5.csv it produced 1,016 rows after SMOTE and reported OSEC accuracy 0.8873, ROC-AUC 0.9516, and Brier score 0.0890, while OVEC reported accuracy 0.8627, ROC-AUC 0.9163, and Brier

score 0.1236. These are protocol-specific reproduction artifacts, not leakage-controlled generalization estimates. The corrected single-split comparison reported OSEC accuracy 0.8957, ROC-AUC 0.9348, and Brier score 0.0875, with stable OSEC accuracy 0.9000, ROC-AUC 0.9360, and Brier score 0.0864; these values are not the primary corrected evidence because they use one split. Accordingly, the original paper's numerical claims are not treated as validated corrected-pipeline results.

IV. Methods

A. Corrected Pipeline

All data-dependent operations are fitted on the training data of the relevant fold. Feature types are declared in the local dataset manifest: nominal and ordinal coded predictors are one-hot encoded, identifiers are excluded, and numeric missingness/outlier treatment is training-only. Unknown external data must provide an explicit feature mapping before the external gate can pass. The boundary flow is summarized in Fig. 1.

B. Stability-Aware Feature Reduction

For feature j and inner resample b, let $r_{j,b}$ denote the combined training-only relevance score and $s_{j,b}$ its selection indicator. Relevance is estimated after splitting each resample into a fitting part and a validation part; permutation importance is evaluated on the latter, which remains inside the outer training boundary. We define

$$\pi_j = \frac{1}{B}\sum_{b=1}^{B} s_{j,b}, \qquad \bar{r}_j = \frac{1}{B}\sum_{b=1}^{B} r_{j,b}. \tag{1}$$

Features are retained when their selection frequency and mean relevance pass the prespecified thresholds. To test whether this improves stability, the selected sets are compared with a single-pass relevance selector over matched repeated stratified training perturbations using the chance-corrected stability statistic of Nogueira et al. [2]. The reported interval is a paired bootstrap interval over repeat-level stability estimates. This is an operational stability rule, not a causal interpretation of feature importance.

C. Uncertainty and Calibration

The project reports Brier score, reliability-bin error, calibration intercept and slope, split-conformal prediction-set coverage, abstention rate, confidence- error AUC, area under the risk-coverage curve (AURC), and selective risk. The confidence-error AUC tests whether confidence ranks correct predictions above errors; it is not a probability of correctness. Conformal coverage is interpreted under its exchangeability assumption [7] and is measured again under any available dataset shift. The implementation uses ten uniform bins for ECE, omits empty bins, and averages the remaining absolute calibration gaps equally. Its AURC value is the arithmetic mean of cumulative error risks over retained-prefix sizes, rather than a numerically integrated curve; this implementation detail limits direct comparison with studies using a different AURC convention.

V. Experimental Protocol

The historical protocol, corrected common-split protocol, and nested-validation protocol are kept separate. Hyperparameter selection is performed only in inner folds. The final internal holdout is not repeatedly consulted. Reporting follows the transparency principles of TRIPOD+AI [8] and the development/evaluation distinctions in PROBAST+AI [9].

For feature-selection stability, five stratified folds were repeated three times. The stable selector was compared with a one-shot selector using the recorded training-fold feature selections. The standalone corrected stability artifacts used eight relevance resamples, whereas the stable selector embedded in the nested and linkage pipelines used six. For selective reliability and the paired stability–reliability analysis, ten random seeds (42–51) used a 56.25% fitting, 18.75% calibration, and 25% evaluation split. Evaluation labels were not used for fitting or calibration. These bounded repeated analyses are exploratory and are not substitutes for a full confirmatory evaluation.

The primary nested endpoint is outer-fold ROC-AUC for the listed candidates; the stability–reliability linkage is an exploratory paired comparison of stable OSEC against corrected OSEC. PR-AUC, Brier score, calibration, feature stability, and selective metrics are secondary. The reported intervals are descriptive bootstrap summaries over the recorded folds, repeat-level stability summaries, or seeds; they are not independent-patient confidence intervals.

As a separate exploratory benchmark, the 70,000-row Kaggle cardio task was evaluated with three shared stratified outer folds, two inner folds, seed 42, and bounded inner searches for Logistic regression, Random Forest, HistGradientBoosting, XGBoost, LightGBM, and CatBoost. This task has a different target and schema and is not pooled with the two CARDIODIAG benchmarks. A fixed one-third out-of-fold ensemble of XGBoost, HistGradientBoosting, and Logistic regression was evaluated on the same outer folds.

A. Multi-source Shift Evaluation

The multi-source experiments used Cleveland, Hungary, Switzerland, and VA Long Beach as training domains. CD-XD-002 used the 13 shared UCI predictors for retrospective leave-one-source-out evaluation. CD-XD-003 and CD-XD-004 used age, sex, and fasting-glucose status with a 20% source-only calibration partition. For the subsequent reliability-layer experiments CD-XD-005 (Duke) and CD-XD-006 (AngioCAD), each source cohort was stratified into disjoint partitions: 70% model fitting, 15% Platt probability calibration, and 15% conformal/OOD threshold calibration. Logistic regression used equal total fit weight per cohort; Platt calibration used equal total weight per cohort. Preprocessing and imputers were fit only on model-fitting rows.

The reliability layer formed binary split-conformal sets from calibrated nonconformity scores $1 - p(y)$, using the

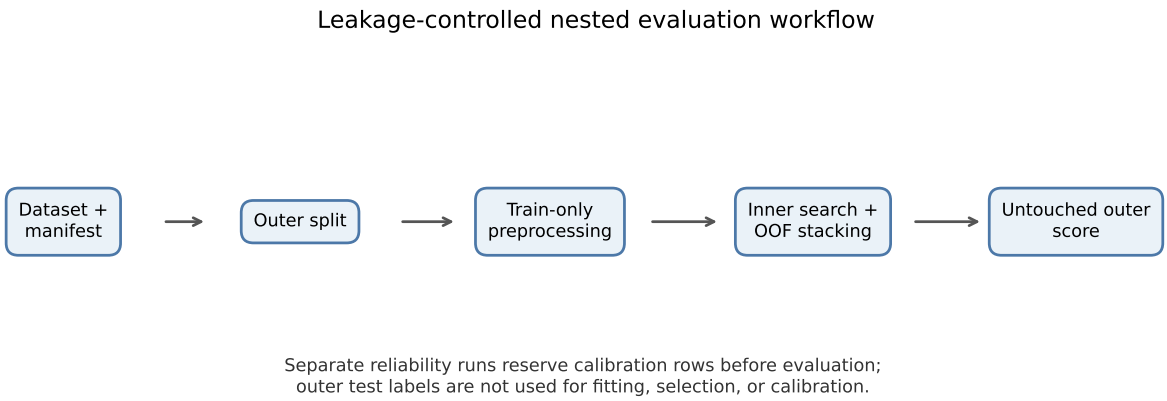


Fig. 1. Leakage-controlled nested evaluation. Preprocessing, model selection, and out-of-fold stacking are confined to the outer training partition; the outer test fold is scored once. Separate reliability experiments reserve calibration data before evaluation.

maximum of the per-source finite-sample 90% quantiles. The feature-support score was mean Euclidean distance to the 10 nearest source-fitting rows, with a per-source 95% calibration threshold. Isolation Forest was retained as an OOD comparator. An inference bundle returned a prediction and probability only for a registered source-domain identifier when the conformal set was singleton and the feature-support score was within threshold. All abstentions withheld the probability, and out-of-support rows also withheld the conformal set; a new or missing domain identifier caused abstention before scoring. The identifier is trusted provenance metadata, not inferred by the model. External labels were used only for scoring. The AngioCAD reliability replay is post hoc because its target had already been examined in CD-XD-004. Target bootstrap intervals use 1,000 class-stratified resamples and condition on observed class counts.

VI. Datasets and Provenance

The supplied Five States and Mendeley datasets are development and replication benchmarks because they were already used in the base study. The Mendeley dataset is attributed to Doppala and Bhattacharyya [10] (DOI: https://doi.org/10.17632/dzz48mvjht.1). Dataset hashes and their assigned roles are recorded in data/manifests/local_datasets.json. The UCI Z-Alizadeh Sani cohort contains 303 patients reported from Shahid Rajaei Cardiovascular, Medical, and Research Center in Tehran [11], [12]; the public AngioCAD dataset reports 413 patients with structured clinical features and artery-level angiographic labels from Shahid Chamran Heart Center in Isfahan [13], [14]; the source cohorts are the four UCI Heart Disease components [15]. The frozen CD-XD-003 protocol maps only age, sex, and fasting glucose to shared definitions. The target labels CAD at $\geq 50\%$ stenosis, whereas the source labels presence at $> 50\%$, and the public files have no patient identifier for direct linkage. The separate- cohort run is therefore a transport stress test, not endpoint-identical external validation. The frozen CD-XD-004 protocol derives the same $> 50\%$ endpoint from 15 artery-segment labels. Only 377 of the 413 labeled AngioCAD IDs had matching clinical-feature rows; these matched rows were used for the one-way evaluation. CD-XD-004 used age, sex, and the binary fasting-glucose threshold > 120 mg/dL. The AngioCAD feature workbook states neither age nor FBS units; the historical protocol assumed age in years and FBS in mg/dL. Values 55–514 are not evidence of a unit. A separate post-hoc FBS-excluded replay is reported, but its age-as-years mapping remains provisional. The 36 unmatched label IDs are recorded, not imputed; their absence reason is not supplied. The negative 26–50 category is a stenosis bin, not an age interval, and prevents isolating exact 50% cases. AngioCAD's site is distinct, but cross-dataset patient-level linkage is unavailable. The data manifests, protocols, split indices, hashes, and compatibility limits are recorded under data/manifests/ and docs/.

The public Mendeley codebook defines only binary disease presence or absence; the diagnostic reference standard and threshold are unspecified. It lists demographics, symptoms, laboratory, ECG, and exercise-test fields, but gives no collection time or intended prediction point for any predictor. The source article maps noofmajorvessels to the number of major vessels seen by fluoroscopy [16]; this is a possible outcome proxy, and its timing is unknown. The identifier patientid is excluded. The remaining variables cannot be established as pre-diagnostic from this release, so Mendeley results are retrospective benchmark replication only. A same-split feature-exclusion sensitivity is reported in Section VII-B.

The Duke ACATH cohort contains 3,504 patients referred for chest pain and is provided by the Vanderbilt Department of Biostatistics from the Duke Cardiovascular Disease Databank [17]. Its sigdz endpoint is significant disease at $\geq 75\%$ diameter narrowing, compared with $> 50\%$ in the UCI source. CD-XD-005 used age, sex, and serum cholesterol; cholesterol was missing for 1,246 patients. Duke sex coding was mapped to the UCI convention. This cohort supplies a source-distinct transport stress test but not endpoint-identical external validation.

TABLE I
Mendeley codebook and predictor-timing audit. No predictor has a documented measurement time relative to the target. The dataset is retained as a retrospective benchmark, not a pre-diagnostic cohort.

Field	Codebook meaning	Provenance / timing assessment	Disposition
patientid	Patient identification number	Administrative identifier; possible row/source encoding	Exclude
age	Age in years	Demographic; collection time and availability at prediction are absent	Retrospective benchmark only
gender	0 female, 1 male	Demographic; collection time absent	Retrospective benchmark only
chestpain	Typical, atypical, non-anginal, asymptomatic	Symptom; relation to diagnosis and index time unknown	Timing unclear
restingBP	Resting blood pressure, mmHg	Setting, treatment state, and measurement time unknown	Timing unclear
serumcholestrol	Serum cholesterol, mg/dL	Sample timing, fasting status, and treatment unknown	Timing unclear
fastingblood sugar	Binary indicator for >120 mg/dL	Measurement time and testing protocol unknown	Timing unclear
restingrelectro	Normal, ST–T abnormality, or probable/definite LVH	ECG-derived diagnostic evidence; timing unknown	Potential outcome-proximal field
maxheartrate	Maximum heart rate achieved	Test context and timing not given	Potentially test-derived; timing unclear
exerciseangia	Exercise-induced angina, 0/1	Exercise-test protocol and timing unknown	Potentially test-derived
oldpeak	Numeric ST-related field, range 0–6.2	Measurement protocol and timing unknown	Potential outcome-proximal field
slope	Peak-exercise ST slope category	Exercise-test-derived; time relative to target unknown	Potentially post-diagnostic
noofmajor vessels	0–3; count seen by fluoroscopy	Angiographic evidence can be downstream of diagnosis	Possible outcome proxy; timing unknown
target	0 absence / 1 presence of heart disease	No reference standard, stenosis threshold, adjudication, or index date supplied	Do not equate to UCI/AngioCAD endpoint

The UCI Statlog (Heart) release was screened as a possible additional cohort. After harmonizing its binary label to the UCI num>0 convention, all 270 Statlog records exactly matched Cleveland on the 13 predictors and outcome; it was therefore excluded from external evaluation rather than counted as a fifth independent source; the source record is available at https://archive.ics.uci.edu/dataset/145/statlog+heart.

VII. Results

Table II reports the post-boundary-corrected nested-validation artifacts. It is internal evidence only and must not be described as independent external evidence. Earlier historical and schema-corrected artifacts remain available under their original prefixes and are not silently replaced.

Figure 2 visualizes pooled outer-fold predictions; its pooled curve scores are distinct from the fold-mean metrics in Table II. Figures 3 and 5 show complementary probability and abstention diagnostics. Figure 4 shows fixed-threshold error counts; Fig. 6 summarizes feature selection frequency, Fig. 7 shows unlabelled distribution differences, Fig. 8 reports exploratory AURA-ML ablations, and Fig. 9 shows synthetic stress tests. None supplies independent validation.

A. Nested Validation

The corrected nested runs used three outer folds, one outer repeat, and three-fold inner selection with ROC-AUC scoring. Corrected OVEC had the highest listed fold-mean ROC-AUC on both datasets: 0.9295 on Five States and 0.9957 on Mendeley. Stable OSEC was 0.9253 versus corrected OSEC 0.9259 on Five States, and 0.9942 versus 0.9943 on Mendeley. The small differences do not establish a uniform predictive advantage. Stable-candidate threshold tuning occurred inside inner folds over 0.5, 0.75, and 1.0; these results therefore are not numerically interchangeable with the fixed-threshold stability experiment.

B. Feature and Harmonization Sensitivity

The Mendeley nested runs were repeated with the same outer and inner splits after excluding noofmajorvessels. Across the five candidates, the fold-mean ROC-AUC changed by only +0.0003 to +0.0007 (Table III). Thus, this one potentially downstream variable does not explain the high internal discrimination. The unspecified target reference standard and the timing of the remaining predictors still prevent an early-detection interpretation.

The AngioCAD feature sensitivity removed FBS from both source fitting and target scoring while retaining the source partitions. This is post hoc because AngioCAD had already been examined. In the seven-subset replay, source-Platt ROC-AUC was 0.788 for age plus sex and 0.787 when FBS was added; the paired candidate-minus-reference AUC difference was -0.0009 (95% conditional interval -0.0060 to 0.0048). Brier score and ECE were 0.176 and 0.196 for age plus sex versus 0.185 and 0.216 for age, sex, and FBS. These comparisons are unadjusted and do not validate the FBS unit. The AngioCAD workbook also does not state the age unit, so age-as-years remains provisional. The separate age/sex Isolation Forest replay had source-versus-target AUROC 0.544 (95% interval 0.496–0.592), consistent with no useful separation. The sensitivity does not establish reliable shift detection (Tables IV and V).

TABLE II
Nested internal validation summary.

Dataset	Candidate	ROC-AUC	PR-AUC	Brier	ECE	ROC-AUC CI
heart5.csv	corrected_logistic	0.928 ± 0.006	0.935	0.107	0.102	[0.907, 0.942]
heart5.csv	corrected_ovec	0.930 ± 0.007	0.933	0.100	0.086	[0.909, 0.944]
heart5.csv	corrected_osec	0.926 ± 0.011	0.928	0.112	0.121	[0.889, 0.930]
heart5.csv	stable_logistic	0.928 ± 0.006	0.935	0.107	0.109	[0.907, 0.942]
heart5.csv	stable_osec	0.925 ± 0.010	0.929	0.109	0.113	[0.896, 0.934]
MendeleyCardiovascular_Disease_Dataset.csv	corrected_logistic	0.993 ± 0.004	0.994	0.027	0.152	[0.988, 0.996]
MendeleyCardiovascular_Disease_Dataset.csv	corrected_ovec	0.996 ± 0.001	0.997	0.025	0.158	[0.992, 0.998]
MendeleyCardiovascular_Disease_Dataset.csv	corrected_osec	0.994 ± 0.004	0.995	0.023	0.200	[0.986, 0.997]
MendeleyCardiovascular_Disease_Dataset.csv	stable_logistic	0.993 ± 0.004	0.994	0.027	0.140	[0.988, 0.996]
MendeleyCardiovascular_Disease_Dataset.csv	stable_osec	0.994 ± 0.004	0.995	0.023	0.242	[0.986, 0.997]

TABLE III
Mendeley nested-validation sensitivity after excluding the potentially post-diagnostic fluoroscopy-derived predictor. Entries are fold-mean ROC-AUC ± fold standard deviation across the same three outer folds; they are descriptive, not patient-level confidence intervals.

Candidate	All predictors	Excluding major vessels	Difference
corrected_logistic	0.9930 ± 0.0036	0.9933 ± 0.0039	+0.0003
corrected_ovec	0.9957 ± 0.0013	0.9961 ± 0.0025	+0.0003
corrected_osec	0.9943 ± 0.0037	0.9949 ± 0.0045	+0.0007
stable_logistic	0.9930 ± 0.0037	0.9933 ± 0.0039	+0.0003
stable_osec	0.9942 ± 0.0037	0.9948 ± 0.0043	+0.0006

TABLE IV
Post-hoc AngioCAD sensitivity to the undocumented FBS unit. Predictive bootstrap intervals condition on the 79 negative and 298 positive matched records. OOD intervals compare source-calibration and target feature scores.

Features	Raw AUC (95% CI)	Calibrated Brier	Calibrated ECE	OOD AUC (95% CI)
Age, sex, FBS	0.787 (0.734–0.838)	0.185	0.216	0.589 (0.541–0.635)
Age, sex only	0.788 (0.734–0.838)	0.176	0.196	0.544 (0.496–0.592)

TABLE V
Post-hoc source-calibrated AngioCAD feature ablation using frozen UCI source partitions. Target-label results are descriptive, unadjusted for seven comparisons, and are not fresh validation. FBS-containing rows assume AngioCAD FBS is measured in mg/dL. The age+sex rows exclude FBS, but the AngioCAD workbook does not state age units, so age-as-years remains provisional. Intervals are class-stratified bootstrap intervals conditional on the previously examined target cohort.

Features	Source-Platt ROC-AUC (95% CI)	Brier	10-bin ECE	Unit status
Age	0.639 (0.579–0.702)	0.190	0.174	Age-as-years assumed
Sex	0.690 (0.628–0.747)	0.194	0.211	Categorical
Age + sex	0.788 (0.734–0.842)	0.176	0.196	FBS excluded; age unit assumed
FBS	0.354 (0.306–0.401)	0.205	0.195	mg/dL assumption
Age + FBS	0.632 (0.570–0.694)	0.199	0.192	mg/dL and age-unit assumptions
Sex + FBS	0.771 (0.717–0.823)	0.202	0.233	mg/dL assumption
Age + sex + FBS	0.787 (0.733–0.840)	0.185	0.216	mg/dL and age-unit assumptions

C. Reliability

The linkage artifacts compare stable OSEC with corrected OSEC over ten seeds. Confidence-error AUC was above 0.5 for both candidates on both benchmarks, but the paired differences did not consistently favor the stable selector. Conformal coverage was measured on the evaluation partition and is not a guarantee under shift. The linkage intervals and rank correlations are descriptive and exploratory.

D. Feature-Selection Stability

The post-boundary-corrected stability comparison is shown in Table VII. The stable-minus-one-shot differences were 0.0974 (95% descriptive bootstrap interval [-0.0272, 0.1978]) on Five States and 0.0687 ([0.0000, 0.1671]) on Mendeley. The Mendeley interval touches zero, and the artifact's prespecified positive-evidence flag is false. These intervals are based on only three repeat-level differences and should be interpreted as limited comparative evidence, not patient-level confidence intervals or proof of generality.

E. Stability–Reliability Linkage

Table VIII tests whether a more stable selected set translated into better held-out reliability on the same repeated splits. Differences are stable OSEC minus corrected OSEC; positive AURC or ECE differences indicate worse selective risk or calibration, respectively. The paired intervals include zero for all listed comparisons.

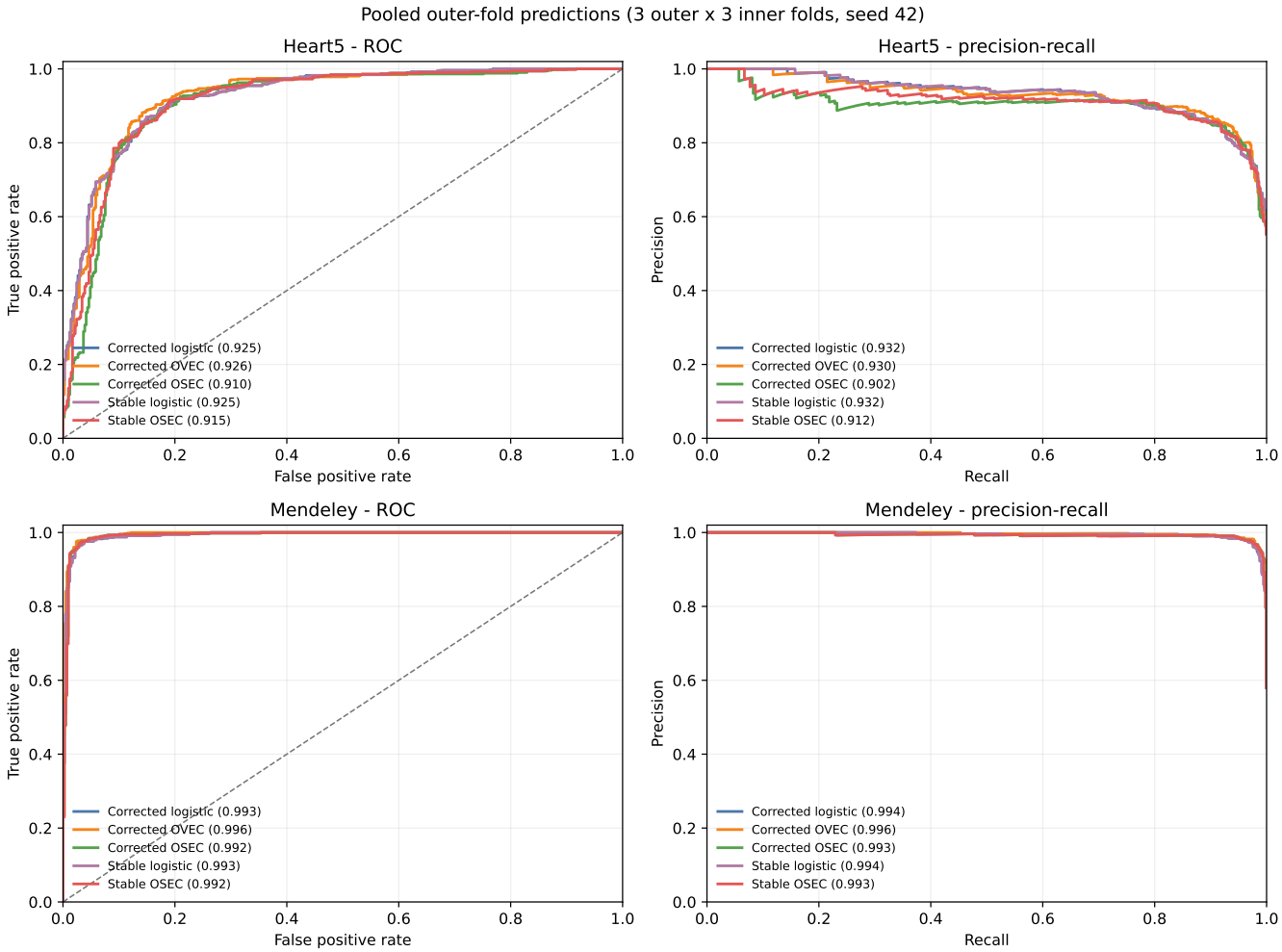


Fig. 2. ROC and precision–recall curves from pooled outer-fold probabilities for the three-outer-by-three-inner, seed-42 nested runs. Legend values are pooled scores; the main result table reports fold means.

TABLE VI
Ten-seed internal selective-reliability summaries from the post-boundary-corrected linkage artifacts. Paired differences and descriptive bootstrap intervals are reported in Table VIII; these are not independent external-validation estimates.

Dataset	Candidate	Conf.-error AUC	AURC	Conformal coverage	ECE
Five States	Corrected OSEC	0.762	0.0505	0.914	0.122
Five States	Stable OSEC	0.760	0.0516	0.915	0.121
Mendeley	Corrected OSEC	0.905	0.0041	0.914	0.221
Mendeley	Stable OSEC	0.905	0.0042	0.914	0.217

On Mendeley, ROC-AUC, AURC, and full-coverage risk favored corrected OSEC, while the other intervals included zero. Thus, selector stability did not act as a consistent surrogate for predictive or selective reliability.

F. Source-Distinct Transport Stress Test

After the source-only protocol was frozen, a single model was trained on all four UCI source cohorts and evaluated once on the source-distinct Z-Alizadeh Sani cohort. The harmonized inputs were age, sex, and fasting-glucose status; the predictor was fixed logistic regression, with equal total fitting weight per source cohort. A Platt calibrator and Isolation Forest warning threshold were fit using source data only. The external cohort had 303 records, including 216 CAD and 87 Normal labels. No target labels were used for fitting, calibration, or threshold selection.

The source-only model achieved ROC-AUC 0.720 (class-stratified bootstrap 95% CI 0.651–0.780) and average precision 0.853 (0.807–0.894). Platt calibration reduced Brier score from 0.200 to 0.193 and 10-bin ECE from 0.136 to 0.110; paired bootstrap intervals for these differences included zero. The OOD score separated source calibration and target rows weakly (AUROC 0.604, 95% CI 0.552–0.652) and flagged 14.9% of target rows at a 5.3% source calibration false-alert rate. Flagged rows had a Brier score

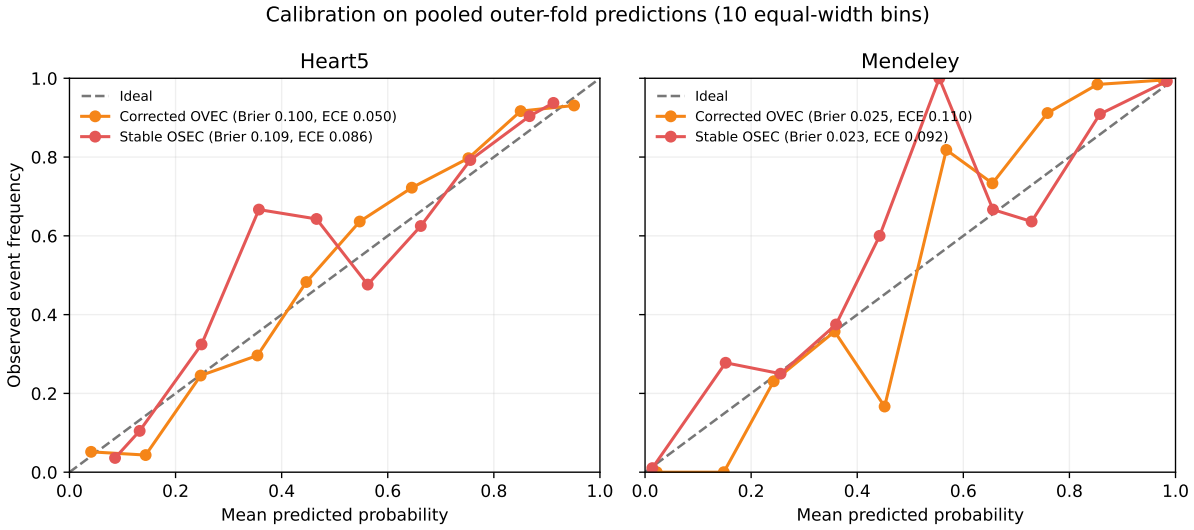


Fig. 3. Ten-bin calibration plots on pooled outer-fold predictions. The Mendeley reliability curves are irregular and ECE is material despite high discrimination, illustrating that ROC-AUC alone does not establish calibrated probabilities.

TABLE VII
Post-boundary-corrected feature-selection stability. Φ is the Nogueira chance-corrected stability statistic; intervals summarize the three repeat-level paired differences and are exploratory.

Dataset	Stable Φ	One-shot Φ	Difference	95% bootstrap CI
Five States	0.112	0.014	0.097	[-0.027, 0.198]
Mendeley	0.284	0.215	0.069	[0.000, 0.167]

TABLE VIII
Paired stability–reliability linkage. Φ is cross-seed selected- set stability. Δ values are stable OSEC minus corrected OSEC means; the full bootstrap intervals are stored in the machine-readable linkage artifacts.

Dataset	Cross-seed Φ	Δ ROC-AUC	Δ Conf.-AUC	Δ AURC	Δ ECE
Five States	0.201	-0.0009	-0.0015	+0.0011	-0.0001
Mendeley	0.178	-0.0002	-0.0001	+0.0002	-0.0045

of 0.238, versus 0.185 in accepted rows, but only five flagged rows were negative. The result does not establish reliable rejection or calibration under shift.

The target site's outcome uses $\geq 50\%$ stenosis, while the source outcome uses $> 50\%$; individual stenosis values are unavailable to reconcile the cut point. The public files also have no patient identifiers for record-level overlap checks. We therefore describe this as a source-distinct transport stress test, not endpoint-identical external validation. The measured cohort is small and single-site, and does not estimate variation across hospitals or time.

G. Endpoint-Aligned External Evaluation

The frozen CD-XD-004 pipeline evaluated AngioCAD after source-only fitting and threshold selection. The analysis used 377 matched records from a separate Isfahan hospital, of which 298 (79.0%) were positive and 79 were negative. The dataset article reports 413 patients, but the released clinical feature workbook lacks rows for 36 labeled IDs. Age and FBS were missing for 6 and 11 matched patients, respectively, and were imputed using source-fit values. The target was positive when any of 15 artery segments had a stenosis category above 50%; this aligns the UCI source endpoint. No cross-dataset person-level linkage was possible.

The source-only predictor achieved ROC-AUC 0.787 (class-stratified bootstrap 95% CI 0.734–0.838) and average precision 0.932 (0.913–0.952). Platt calibration did not change ranking. It reduced Brier score from 0.190 to 0.185 and log loss from 0.569 to 0.548 by point estimate, while 10-bin ECE increased from 0.211 to 0.216. Paired bootstrap intervals for the Brier-score change (-0.0133 to 0.0049), log-loss change (-0.0415 to 0.0006), and ECE change (-0.0213 to 0.0164) included no difference. Mean predicted risk after calibration was 0.574, below the observed 0.790 positive fraction; the calibration intercept and slope were 1.219 and 1.436. Thus the small Brier and log-loss improvements do not establish adequate target calibration.

The OOD detector flagged 44/377 (11.7%) target records, with 88.3% accepted coverage and a 5.3% source-calibration false-alert rate. Source-versus-target OOD AU-

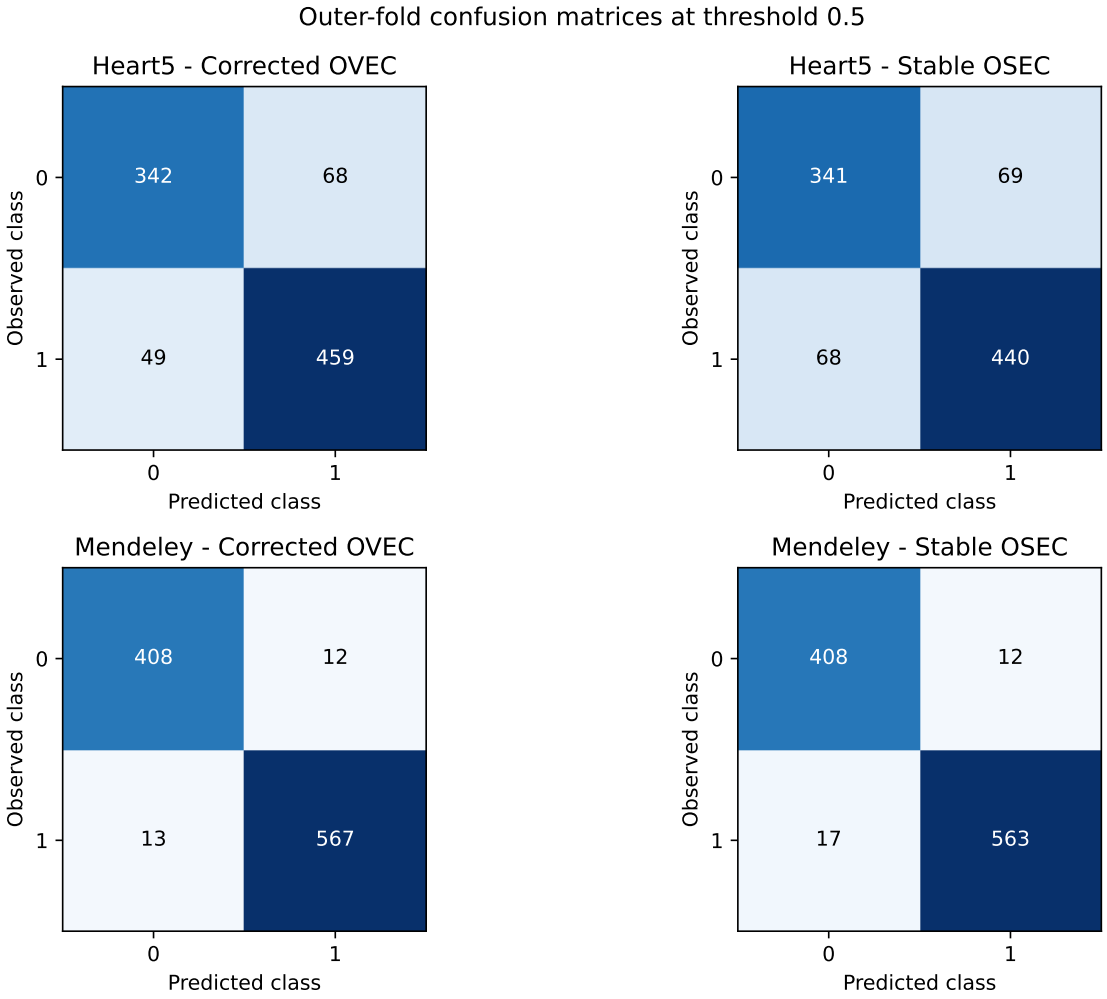


Fig. 4. Pooled outer-fold confusion matrices at a fixed 0.5 probability threshold. Threshold-specific counts are descriptive and are not a substitute for the threshold selection protocol or a clinical operating point.

TABLE IX
Source-only transfer results on the Z-Alizadeh Sani cohort. Bootstrap intervals condition on the observed target class counts.

Metric	Estimate	95% interval
ROC-AUC	0.720	0.651–0.780
Average precision	0.853	0.807–0.894
Brier, raw / calibrated	0.200 / 0.193	0.192–0.208 / 0.174–0.213
ECE, raw / calibrated	0.136 / 0.110	0.115–0.172 / 0.097–0.154
OOD AUROC	0.604	0.552–0.652

ROC was 0.589 (95% CI 0.541–0.635), indicating weak separation. The calibrated Brier score was 0.179 among accepted records and 0.233 among the 44 flagged records. This descriptive difference is not evidence that the warning reliably identifies individual errors.

H. Source-Domain Abstention and Operating Boundary

CD-XD-006 re-ran the endpoint-aligned AngioCAD transfer with disjoint probability and conformal-calibration partitions. The source-only predictor achieved ROC-AUC 0.790 (95% class-stratified bootstrap CI 0.733–0.837). Platt calibration changed Brier score from 0.189 to 0.174 and 10-bin ECE from 0.211 to 0.192. The 90% source-domain conformal set covered all 377 target records empirically, but only 33 (8.8%) received singleton sets. After combining singleton-set and feature-support rules, only 21 (5.6%) received automatic outputs; no errors were observed, but all 21 accepted records were positive. This small one-class subset does not establish low error at a useful operating coverage. The kNN feature support detector separated target from source calibration rows weakly (AUROC 0.580, 95% CI 0.527–0.635).

The metadata-based operating-domain gate withheld probabilities for all 377 AngioCAD records because the site

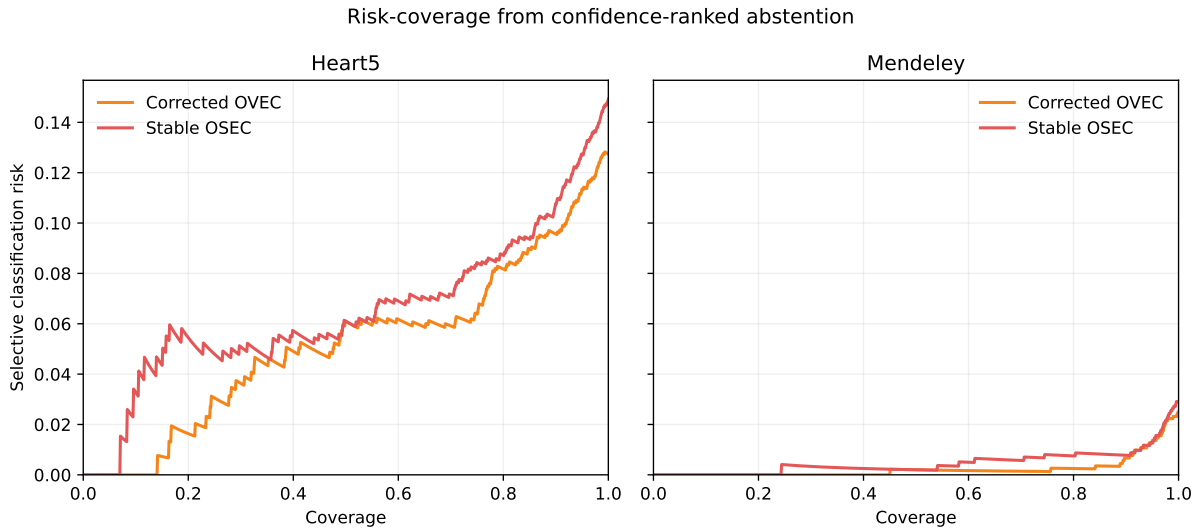


Fig. 5. Selective classification risk as lower-confidence predictions are abstained from. Curves describe internal pooled outer-fold predictions and do not certify safety or clinical benefit.

TABLE X
Source-only transfer results on the endpoint-aligned AngioCAD cohort. Bootstrap intervals condition on the 79 negative and 298 positive matched records.

Metric	Raw	Source Platt calibrated
ROC-AUC	0.787 (0.734–0.838)	0.787 (0.734–0.838)
Average precision	0.932 (0.913–0.952)	0.932 (0.913–0.952)
Brier score	0.190 (0.184–0.196)	0.185 (0.171–0.200)
Log loss	0.569 (0.556–0.581)	0.548 (0.516–0.580)
10-bin ECE	0.211 (0.203–0.232)	0.216 (0.198–0.234)
Balanced accuracy at 0.5	0.630	0.719
Sensitivity at 0.5	0.906	0.728
Specificity at 0.5	0.354	0.709

is not in the UCI source registry. This gate is deterministic when trusted site metadata is provided; it is not a feature-based estimate of domain membership. CD-XD-006 is a post-hoc policy replay because AngioCAD outcomes had already been examined in CD-XD-004.

CD-XD-005 evaluated the same abstention method on the newly sourced Duke cohort, using age, sex, and cholesterol. ROC-AUC was 0.661 (95% bootstrap CI 0.642–0.680). Source Platt calibration worsened Brier score from 0.216 to 0.241 and ECE from 0.119 to 0.198. The kNN and Isolation Forest OOD AUROCs were 0.556 and 0.598, respectively. The conformal set covered 98.3% of Duke records empirically but was singleton for 13.9%; the combined method automatically classified 401 (11.4%) with 41 errors (10.2%). The site gate withheld probabilities for all 3,504 records. Because Duke's endpoint begins at 75% stenosis, these differences combine cohort, endpoint, and measurement shift and are not a direct test of same-endpoint generalization.

An independent post-hoc component audit recalculated coverage and accepted errors for the frozen probability, conformal, kNN, Isolation Forest, combined, and site-gate policies. The detector-only rules retain 93.4–99.7% coverage on AngioCAD while leaving the fixed-threshold error rate essentially unchanged. The registered conformal+kNN combination covers only 5.6% on AngioCAD and 11.4% on Duke. The unknown-site registry gate remains at 0% coverage for both cohorts. Component results are descriptive and did not select the deployed policy (Table XI).

I. External Validation and Threats to Validity

The Z-Alizadeh Sani run is source-distinct at the reported hospital level, but its threshold includes exactly 50% stenosis and it remains a transport stress test. AngioCAD provides one endpoint-aligned external cohort, but only 377 of 413 labeled patients had matching feature rows, and the public data do not support cross-dataset patient-level linkage. The supplied Mendeley dataset was used in the base paper, and the public UCI component sources overlap the Five States composite. The AngioCAD results show moderate-to-good ranking but poor calibration and weak feature-only OOD separation. The site gate withholds scores for unregistered cohorts, but its 0% automatic coverage on both evaluated target sites is a limitation in utility, not evidence of a successful predictive system. Duke ACATH adds a source-distinct stress test, but its $\geq$ 75% endpoint differs from the UCI $>$ 50%

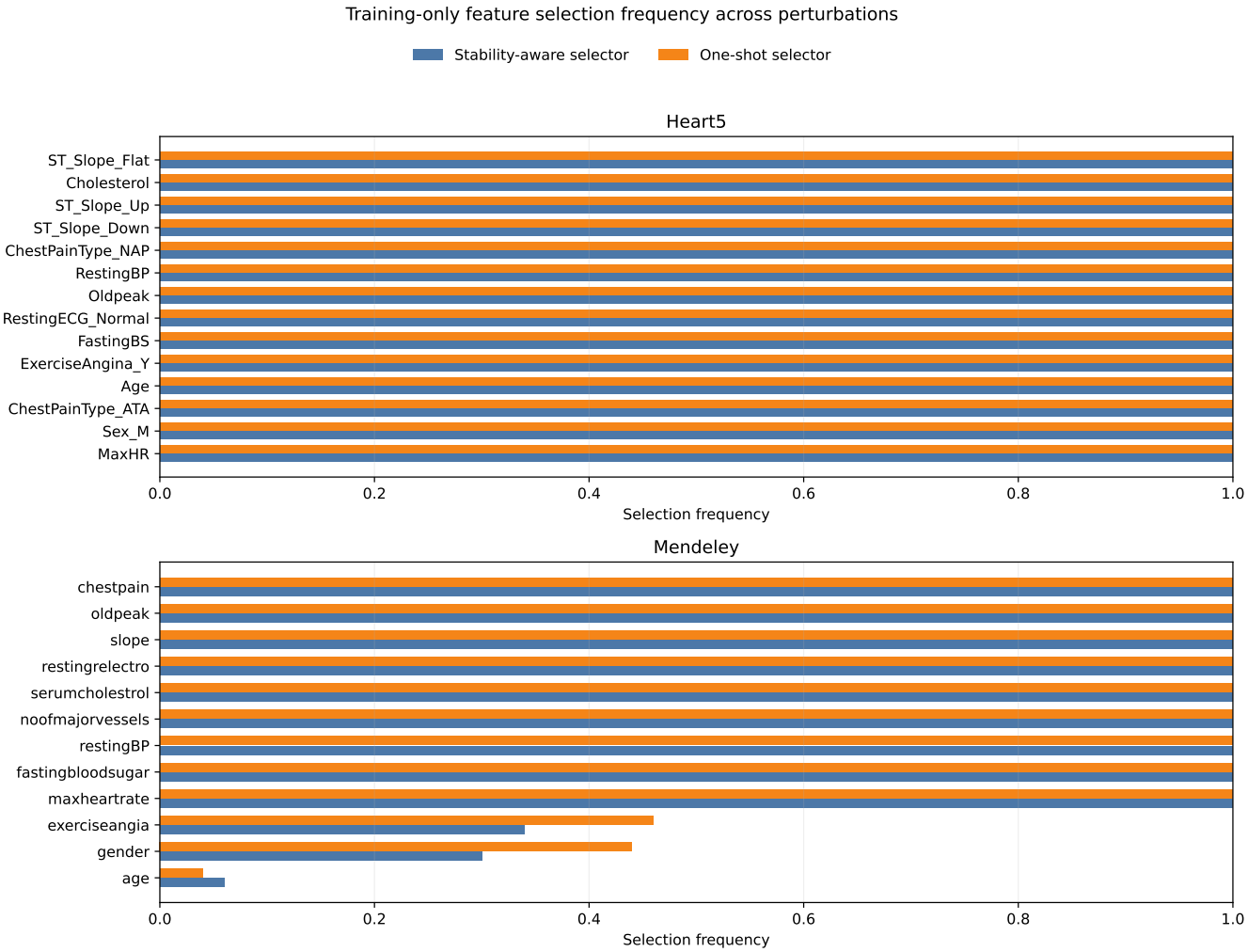


Fig. 6. Training-only feature selection frequencies under the stored perturbation analysis. These frequencies measure feature-set stability; they are not LIME explanation stability or evidence of causal importance.

TABLE XI
Post-hoc comparison of frozen selective-policy components. AngioCAD was previously examined; Duke uses a different endpoint. The domain registry blocks both unregistered cohorts. Target labels are evaluation-only and did not select thresholds or policies.

Policy	AngioCAD coverage	AngioCAD errors	Duke coverage	Duke errors
Fixed probability 0.5, no abstention	100.0%	96/377 (25.5%)	100.0%	1,358/3,504 (38.8%)
Conformal singleton only	8.8%	0/33; one class	13.9%	61/487 (12.5%)
kNN detector only	93.4%	93/352 (26.4%)	95.6%	1,305/3,349 (39.0%)
Isolation Forest only	99.7%	96/376 (25.5%)	97.3%	1,330/3,409 (39.0%)
Registered conformal + kNN	5.6%	0/21; one class	11.4%	41/401 (10.2%)
Unknown-site registry gate	0.0%	No score emitted	0.0%	No score emitted

endpoint. AngioCAD's CD-XD-006 labels were previously inspected, so this replay is not a pristine final test. Additional endpoint-compatible hospitals and temporal or prospective evaluation remain necessary before making broad generalization claims. This caution is consistent with documented heterogeneity and performance degradation when cardiovascular prediction models are evaluated outside their development data [18].

J. Statistical Analysis and Robustness

The study prespecifies ROC-AUC as the primary comparison and reports PR-AUC, classification metrics, Brier score, calibration intercept and slope, uniform ten-bin ECE, selected-set stability, confidence-error AUC, AURC, and conformal coverage as secondary quantities. Paired comparisons use identical partitions for the candidates. No multiplicity-adjusted confirmatory claim is made across the secondary metrics. Synthetic missingness or measurement perturbations, when run, will be reported as stress tests and not as independent validation.

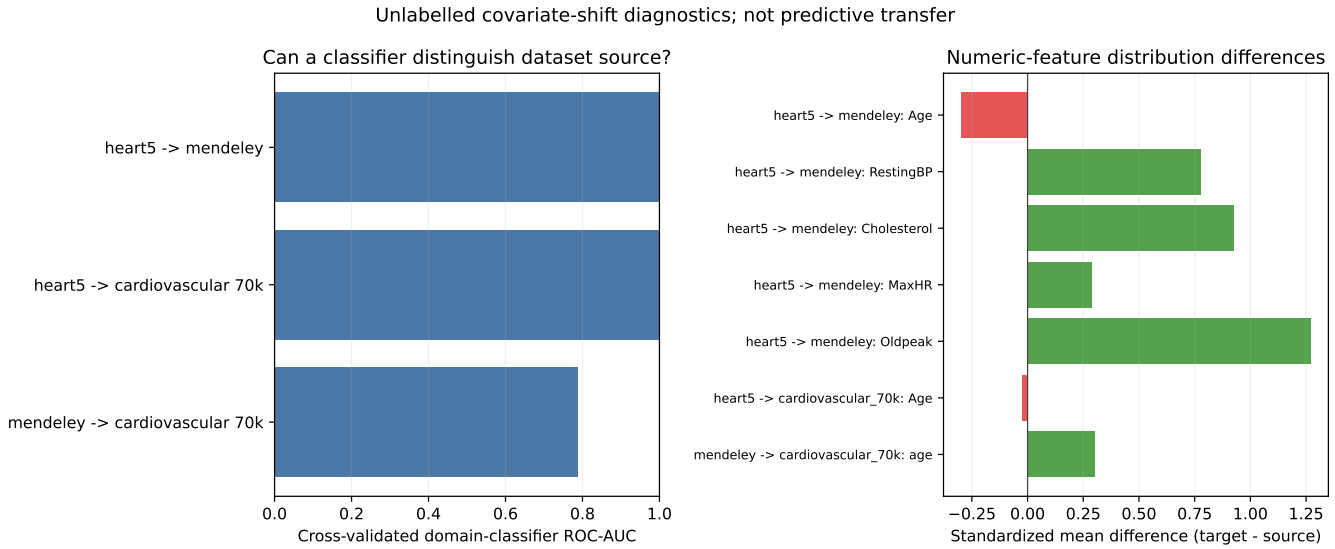


Fig. 7. Unlabelled distribution diagnostics across local datasets. Domain classifier discrimination and standardized mean differences indicate covariate differences only; endpoints are not semantically aligned and no predictive transfer is evaluated.

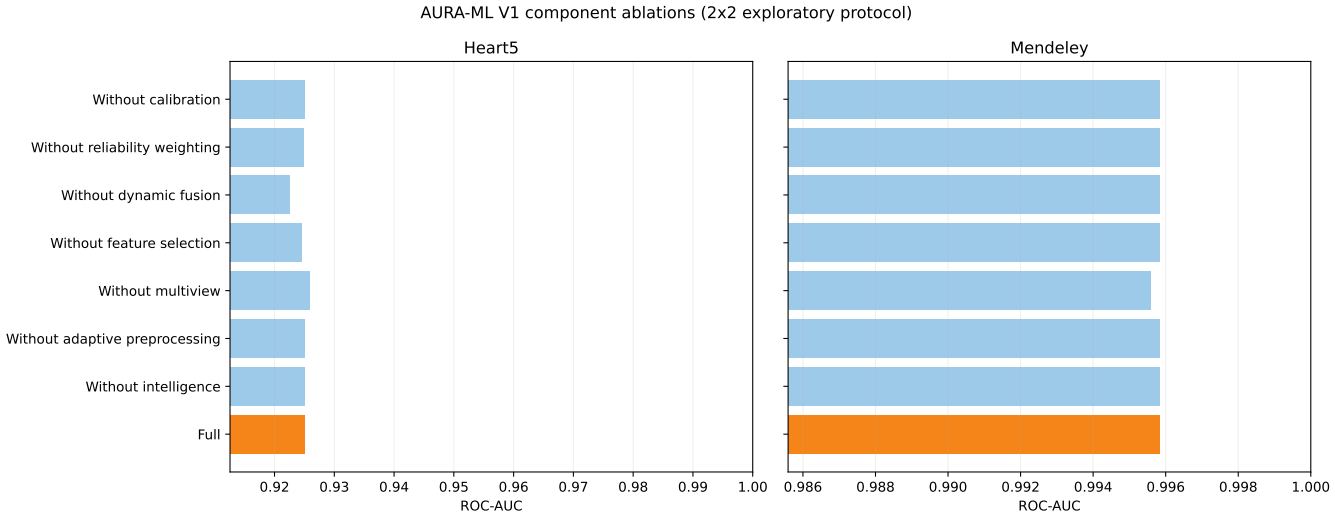


Fig. 8. Exploratory AURA-ML V1 component ablations under the stored 2-by-2 protocol. These bounded results motivate a dataset-conditional interpretation and do not establish an improved general algorithm.

The post-boundary-corrected stability runs found differences of 0.0974 [-0.0272, 0.1978] on Five States and 0.0687 [0.0000, 0.1671] on Mendeley. The ten-seed linkage runs found no consistent reliability advantage: the paired ROC-AUC difference was -0.0009 [-0.0027, 0.0007] on Five States and -0.0002 [-0.0004, -0.0001] on Mendeley, while calibration and selective metrics did not uniformly favor the stable candidate. These are bounded internal results and remain subject to dependence, small effective replication counts, and multiple comparisons.

The separate Kaggle benchmark produced pooled outer ROC-AUC values of 0.80043 for CatBoost and 0.80022 for XGBoost in the tuned run. The prior XGBoost benchmark was 0.80038. The fixed uniform ensemble achieved 0.79897 and therefore did not improve the standalone boosted models. The CatBoost difference is marginal, exploratory, and not evidence of practical or statistical superiority.

A compact candidate-pool rerun evaluated Elastic Net, RBF SVM, KNN, Extra Trees, Gaussian NB, and LDA alongside the scalable boosting candidates. On 70,000 rows, RBF SVM and KNN were prespecified as computationally infeasible and were not run. In that separate protocol, stacking achieved pooled ROC-AUC 0.80028, XGBoost 0.80008, CatBoost 0.79998, and fixed soft voting 0.79943. The small differences do not establish an ensemble or model superiority.

An evaluation-only stress test at seed 42 applied 10% independent missingness and numeric noise with standard deviation equal to 5% of each training feature standard deviation. On Mendeley, ROC-AUC fell from 0.998 to 0.810 under the numeric-noise perturbation for corrected OSEC and from 0.998 to 0.815 for stable OSEC. On

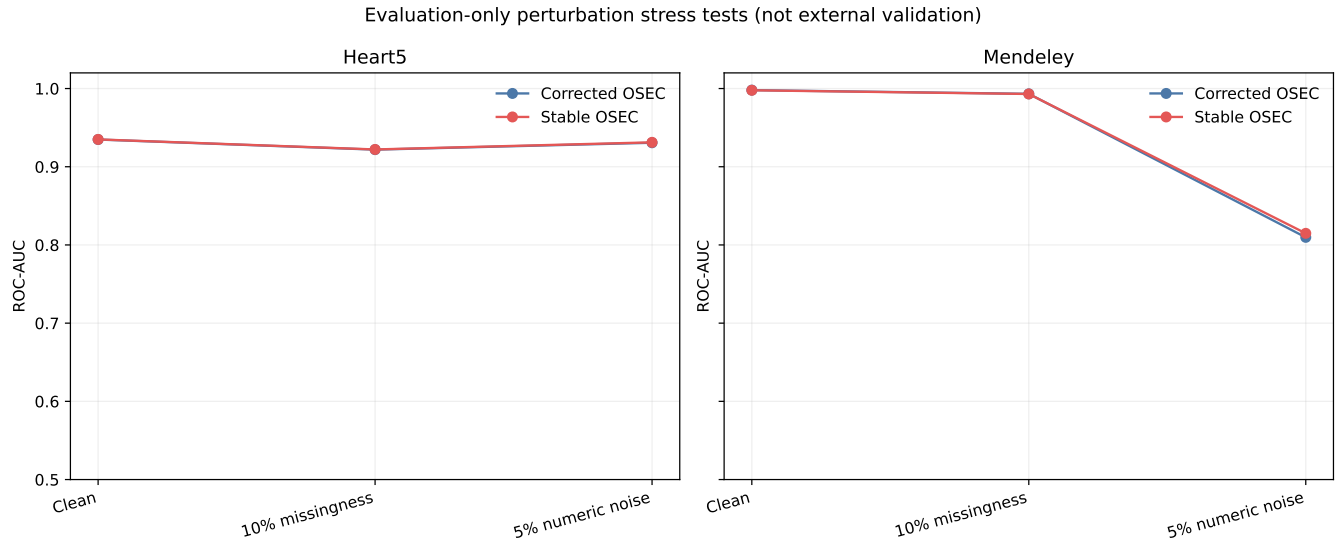


Fig. 9. Evaluation-only synthetic missingness and numeric-noise stress tests. The perturbations are not external data and should not be interpreted as population transport estimates.

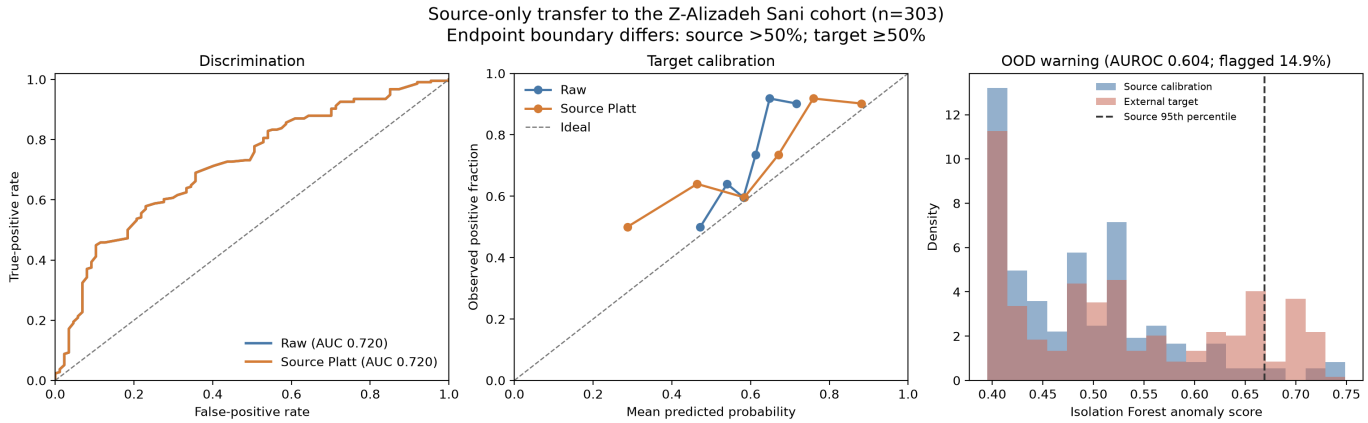


Fig. 10. Source-only transfer results for the Z-Alizadeh Sani cohort. The calibration and OOD components were fit using source data only. The outcome boundary differs at exactly 50% stenosis, so this is a transport stress test.

Five States, 10% missingness reduced ROC-AUC from approximately 0.935 to 0.922 for both candidates. These are synthetic single-split stress tests, not independent external validation.

VIII. Discussion

The nested results do not support a uniform performance advantage for the stability-aware selector. On the Five States benchmark, corrected OVEC has the highest mean ROC-AUC among the listed candidates, while stable OSEC is lower. On the Mendeley benchmark, corrected OVEC and stable OSEC are similar. The post-boundary-corrected stability comparison does not establish a positive stability advantage: its Five States interval crosses zero and its Mendeley interval touches zero. The ten-seed reliability results do not show consistent improvement across discrimination, selective risk, coverage, and calibration. These findings separate feature-set stability from predictive reliability: more repeatable feature selection does not by itself establish better calibrated or safer predictions. A selector is not justified merely because it reduces dimensionality or includes an explainability component.

The transport evidence is mixed. The post-hoc AngioCAD replay ranked patients reasonably well, but automatic coverage was only 5.6%; the feature-support detector was weak, and the registry gate withheld every target score. The fresh Duke cohort showed lower ranking performance, worse calibration after source-only Platt scaling, and a 10.2% error rate among the small subset automatically classified. Its stricter stenosis endpoint prevents attributing that result to dataset shift alone. Together, these results demonstrate a working fail-closed prototype and expose its present operating boundary; they do not establish useful predictive coverage on unseen hospitals.

TABLE XII
Source-only transfer and abstention results. AngioCAD was previously examined; Duke uses a different stenosis threshold. The registry gate withholds scores for both unregistered target sites.

Metric	AngioCAD ($n = 377$)	Duke ACATH ($n = 3{,}504$)
Endpoint	>50% aligned	≥75% mismatch
ROC-AUC (95% bootstrap CI)	0.790 (0.733–0.837)	0.661 (0.642–0.680)
Brier score (raw / Platt)	0.189 / 0.174	0.216 / 0.241
10-bin ECE (raw / Platt)	0.211 / 0.192	0.119 / 0.198
90% conformal set coverage	100.0%	98.3%
Singleton-set coverage	8.8%	13.9%
Combined automatic coverage	5.6%	11.4%
Errors among automatic outputs	0/21	41/401 (10.2%)
OOD AUROC (kNN / IF)	0.580 / 0.579	0.556 / 0.598
Registry-gate automatic coverage	0.0%	0.0%

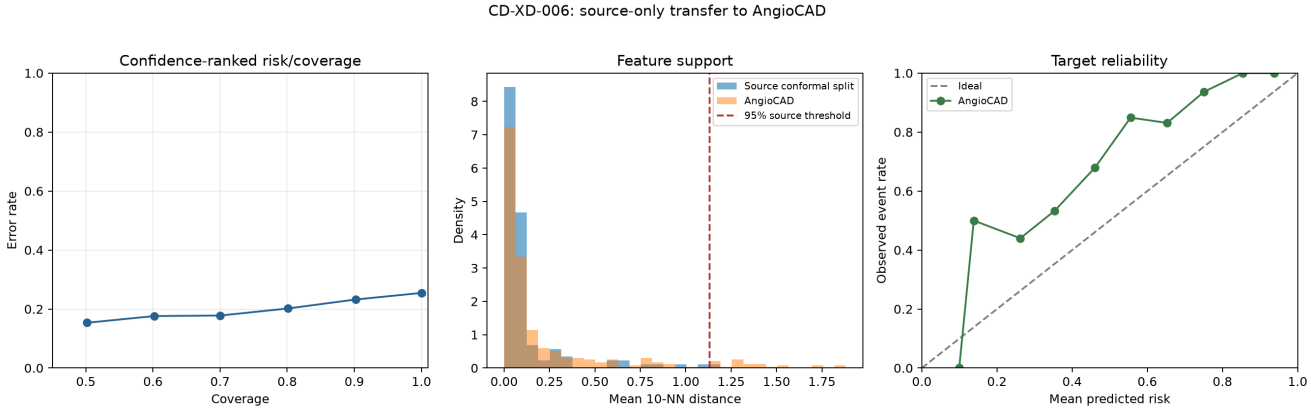


Fig. 11. CD-XD-006 post-hoc AngioCAD replay. The source-calibrated model's feature-support distributions overlap, and confidence ranking does not remove the need for the source-domain gate. All AngioCAD rows are withheld by that gate because the target site is unregistered.

IX. Limitations and Ethical Considerations

This is a retrospective engineering evaluation using public benchmarks; it does not estimate prospective clinical performance. Dataset provenance, license conditions, outcome definitions, prediction timing, and possible source overlap constrain interpretation and transportability. In particular, the Mendeley endpoint has no documented diagnostic reference standard, and the AngioCAD unit sensitivity is post hoc. The study does not establish broad generalization or reliable feature-only shift recognition. The full five-candidate custom nested protocol was not completed on the 70,000-row Kaggle task because of computational cost; a reduced corrected-logistic versus stable-logistic analysis is labeled separately. The source-domain gate relies on trusted site metadata and withheld all target predictions; the feature-only OOD score did not reliably identify either unseen cohort. The conformal sets were calibrated on source partitions, and external empirical coverage does not imply a distribution-free guarantee under dataset shift.

X. Reproducibility

The code, tests, dependency lock, data manifests, protocol reports, and result generators are maintained in the repository. Raw restricted data must not be committed.

XI. Conclusion

An exploratory AURA-ML V2 extension was subsequently evaluated under three outer and three inner folds on seeds 42–44 for the two development benchmarks. The V2 policy removed feature selection after its ablation evidence failed to show a consistent benefit, and compared training-side sigmoid and isotonic calibration and threshold selection. Seed-aggregated ROC-AUC was 0.9244 for AURA-ML V2 versus 0.9257 for corrected OVEC on Five States, and 0.9977 versus 0.9955 on Mendeley. These results are supplementary exploratory evidence, not external validation: the 70,000-row V2 3-by-3 attempt exceeded the local runtime budget without producing a valid artifact, and the Mendeley benchmark was already used in the base study.

This work delivers a reproducible leakage-controlled audit and a falsifiable stability–reliability decoupling audit for CARDIODIAG. Internal evidence shows that feature-set stability and selective reliability can move independently: the corrected stability comparison does not establish a stable-selector advantage, and the stable selector does not produce a consistent predictive or reliability benefit. The endpoint-aligned AngioCAD studies achieved ROC-AUC 0.787 and 0.790, but the latter is a post-hoc replay on a previously examined target. Disjoint source calibration modestly improved point estimates for

Brier score and ECE on AngioCAD, while conformal and feature-support rules automatically covered only 5.6% of its rows. The new registry gate withheld all predictions on unregistered target sites, including AngioCAD and Duke. The Duke stress test had a different outcome threshold and source-only Platt calibration worsened its Brier score. Feature-only OOD separation remained weak. These findings support a reproducible fail-closed research prototype and document its low utility and dataset limitations; they do not establish dependable generalization, clinical readiness, or a novel prediction algorithm. For the stated objective of dependable, uncertainty-calibrated prediction on unseen datasets, this is a research no-go: the available evidence fails the claim. The engineering prototype is complete as a reproducible negative result; reversing this decision requires untouched, endpoint-compatible external data.

Author and Publication Declarations

Ethics and data use: Not determined. This study analyzes secondary public datasets. The project team must verify dataset access and reuse terms and obtain an institutional determination of whether ethics review or an exemption is required. No approval or exemption is claimed in this draft.

Funding: Not determined; confirmation from all authors is pending.

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Competing interests: Not determined; all authors must disclose any relevant interests before submission.

Data availability: Dataset sources, mappings, and input hashes are recorded in the project manifests. Access and reuse terms are dataset-specific. The Kaggle 70,000-record dataset's license is listed as unknown; it and its derived results are not cleared for redistribution. A future public release requires a dataset-by-dataset rights review.

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